The World Model Remembers, the Actor Forgets: Dream Rehearsal for Continual Model-Based RL

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Abstract

Model-based reinforcement-learning agents of the DreamerV3 family forget catastrophically when trained on task sequences, even when an unbounded replay buffer preserves every earlier experience. We ask a question the continual-RL literature has assumed an answer to but never measured: *which component forgets?* Under never-clear replay, pre-registered component-level probes (n=3 seeds throughout) show that the world model retains essentially everything measurable about old tasks — reward discrimination (retention ratio $\approx$ 1.0), value estimates, and termination structure — while the actor's behavior collapses. Forgetting in this regime is a channel problem, not a memory problem. We demonstrate this by intervention: with the world model frozen and identical imagined rollouts, reinforcement learning in imagination fails to recover a lost skill (0/3 seeds), while supervised self-imitation on the world model's own *graded dreams* recovers it on 3/3 seeds with zero environment interaction. Interleaved during training, this *graded dream rehearsal* yields a task-label-free, parameter-constant continual learner: 3/3 four-task chains retained where plain replay passes 0/3, 3/3 eight-task chains, and consistent gains over matched real-episode cloning (paired difference +0.13, bootstrap 95% CI $[0.07, 0.24]$, complete seed separation). The dream-grading step is load-bearing: we characterize two scoring failure modes, provide an offline selection gauge that caught both before they contaminated results, and give a realized-first grading rule that closes them. All experiments were pre-registered with committed protocols; every refuted hypothesis is reported.

Code, pre-registration trail, and run data: `https://github.com/gurpnijjer/dream-rehearsal`
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1 Introduction

An agent that learns task B by destroying its competence at task A cannot learn continually, and reinforcement-learning agents do exactly this. The standard remedies treat the network as the thing to protect: regularize its parameters (EWC [8] and successors), grow it (progressive networks [9], per-task heads), or replay its past data (CLEAR [7]; reservoir buffers). In model-based RL the replay remedy has a natural home — the world model trains on buffered experience — and the resulting recipe, a Dreamer-class agent with a never-clear buffer, is the field's current working answer [1, 2, 3].

That line of work carries an assumption stated explicitly but, to our knowledge, never tested. WMAR [2] writes: "We hypothesise that maintaining the world model's accuracy across different environments will help preserve performance." ARROW [3] infers world-model fidelity backward from behavioral metrics. The premise — protect the world model and the policy will follow — has not been checked at the component level, because the standard evaluation measures only episodic return, which cannot distinguish a forgotten representation from a forgotten behavior.

We checked. The first half of the premise is right: under never-clear replay the world model retains essentially everything we can measure about old tasks. The second half is wrong: the actor forgets anyway. The training signal for old tasks — accurate rewards, calibrated values, intact dynamics over old-task states — is present inside the agent's own imagination throughout training, and the policy decays regardless. Forgetting, in this regime, is not a memory problem. It is a channel problem: policy-gradient learning fails to convert an intact signal into retained behavior, and we show a supervised channel converts the same signal successfully.

A note on scope, stated here rather than left to the limitations: everything in this paper is established on MiniGrid chains with a 17M-parameter agent. We chose this scale deliberately — it is the scale at which every component can be probed, frozen, checksummed, and re-read per-seed, which is what a localization claim requires. The contribution is the phenomenon, its component-level localization, and the mechanism that exploits it; whether the pattern persists in larger domains is an open empirical question we are actively pursuing and do not claim here.

This paper makes four contributions, in decreasing order of claimed novelty:

1. **A component-level localization of forgetting** in world-model agents (§4): representations, reward heads, and critics survive under replay; only the actor's behavior decays. This contradicts the untested premise underlying replay-based continual MBRL and reframes what needs fixing.
2. **A channel-isolation result** (§5): with a frozen world model and identical imagined data, RL-in-imagination fails to recover a lost skill (0/3 seeds) where supervised self-imitation succeeds (3/3, zero environment steps). The instability is the policy-gradient channel itself.
3. **Graded dream rehearsal** (§6): interleaved supervised self-imitation on world-model-graded imagined rollouts of prior tasks during sequential training. One actor, no task labels at any time, no parameter growth, $\sim$15% compute overhead at four tasks. It converts the isolation upper bound into a mechanism — 3/3 four-task and 3/3 eight-task chains retained, against 0/3 for the plain-replay recipe — and consistently outperforms matched real-episode cloning.
4. **Grading is load-bearing** (§7): we characterize two failure modes of imagined-trajectory scoring (post-terminal contamination; promises outbidding realized successes), give a realized-first rule that closes both — an imagination-safe port of the self-imitation advantage gate [5] — and introduce an offline gauge that measures selection quality directly, which caught both failure modes before they cost environment interaction.

Every experiment was pre-registered before running — protocols, pass bars, and interpretation matrices committed to version control, with off-machine timestamped archives — and refuted hypotheses are reported in the main text, including two scoring bugs our own gauges caught. We consider the gauges part of the method (§7), and the transparency part of the evidence.

2 Related Work

Replay-protected continual MBRL. Continual-Dreamer [1], WMAR [2], and ARROW [3] form the line whose premise §4 measures: a Dreamer-family agent whose world model trains on a retained (plain, augmented, or distribution-matched) replay buffer, with continual competence attributed to preserved world-model accuracy. Our plain-replay baseline is Continual-Dreamer's recipe at our scale. ARROW is the closest concurrent method in setting, and it is orthogonal by our own localization: ARROW protects the world model's training distribution with distribution-matched *real* replay, and §4 shows that under never-clear replay the world model is already the protected component. The actor is the component that forgets, and no method in this line rehearses the actor from imagination.

Generative replay and pseudo-rehearsal. Deep Generative Replay [10] is the ancestral idea: train a generator of past data and rehearse from it. Dream rehearsal is imagination-native — no generator is trained, because the world model already in the loop *is* the generator — and it grades trajectories rather than replaying them. ReGen [4] (June 2026, concurrent) is structurally the closest published mechanism: world-model pseudo-replay plus a single policy trained by behavior cloning, in continual *imitation* learning on a video-diffusion model with instruction-conditioned replay. It is neither task-agnostic nor RL; we cite it as convergent evidence for the mechanism family.

Self-imitation. The advantage gate of Self-Imitation Learning [5] — imitate a trajectory only when its realized return beats the critic's estimate — is prior art for §7's grading rule on real data. Our contribution is scoped to making that gate safe for *imagined* data, where the critic's optimism is unconstrained by real outcomes and where dreams do not stop at their own endings (§7).

Apparent counter-evidence. Wołczyk et al. [6] show that model-free fine-tuning irreversibly overwrites pre-trained capabilities. We reproduce the distinction rather than contradict it: in their regime no mechanism exists that could preserve anything (no world model, no dynamics objective, no retained data), and we observe total representation overwrite in exactly that condition (§4.2, no-replay bracket). Our claim is regime-specific to replay-maintained world models — the regime the continual-MBRL literature actually operates in. The plasticity literature — capacity loss under nonstationarity [12], primacy bias [13] — documents degradation of the RL channel during ordinary training and is mechanistically adjacent to §5's channel finding; it does not, however, measure backward transfer against an intact world model, which is the comparison our setting isolates.

Parameter isolation. Frozen per-task modules with a task selector are the classical isolation approach; §4.4 constructs the task-agnostic version (frozen actor snapshots plus a task router) as a measured reference point and shows why storing policies caps retention at snapshot quality. CLEAR [7] is the real-data ancestor of §6.2's cloning comparison, which we distinguish empirically rather than rhetorically.

3 Experimental Setup

Agent. DreamerV3 [11] (PyTorch reference implementation; 17M-parameter world model: CNN encoder, RSSM, reward/continuation/decoder heads; 1.8M-parameter actor; imagination horizon H=15). Greedy exploration; all hyperparameters are the reference MiniGrid configuration unless stated (Appendix A).

Tasks and chains. MiniGrid [14], 64×64 RGB observations. Core chain (§§4–7): DoorKey-5x5 → SimpleCrossingS9N1 → LavaGapS5 → MultiRoom-N2-S4 — mixed episode horizons (~10 to ~80 policy steps) and one lethal task, properties that turn out to matter (§7). The scale-up chain (§8) appends four audition-gated tasks: LavaCrossingS9N1, DistShift2, DoorKey-6x6, Unlock (eight tasks, six distinct mechanics, three lethal).

Protocol. Sequential phases. During phase i the agent acts only in task i; a shared never-clear buffer accumulates all experience (the Continual-Dreamer recipe at our scale). Every 2,000 environment steps, every learned task is evaluated for 10 real-environment episodes. A phase advances when all learned tasks score ≥0.6 (mean return) for 3 consecutive evaluation rounds, or at a 150k-step cap. Final retention is the last-3-round mean per task; a seed passes if all tasks finish ≥0.6. n=3 seeds everywhere; no per-seed tuning; every experiment's bars and interpretation matrix were committed before launch. Because run-level nondeterminism is substantial (the same seed can produce qualitatively different trajectories), we report per-seed values and dispersion rather than pass counts alone.

4 Which Component Forgets?

4.1 The phenomenon

Two tasks, no replay: retention of task A after training task B falls from 0.96 to 0.0/0.27/0.12 across seeds. Four tasks *with* unbounded replay: 0/3 chains pass, and the casualty is the same task on every seed — SimpleCrossing retains 0.35/0.51/0.25 while DoorKey (~0.96), LavaGap (~0.94), and MultiRoom (0.79–0.82) survive. Replay protects some tasks and not others; whatever replay protects, it is not uniformly "the policy."

4.2 Freeze bracket

Freezing the actor's weights at task-A competence while the world model continues training: retention 0.0/0.0/0.07 — actor protection without representation stability is useless; the latent space moves under the frozen head. Freezing the world model instead: the new task never learns at all (the actor learns exclusively in imagination, and a frozen world model only dreams task A) — one seed, terminated early as a foregone conclusion and reported as direction-finding only. The bracket eliminates both naive protection strategies and sharpens the question: with replay keeping the world model trained on all tasks, what exactly is still being lost?

4.3 The world model retains everything we can measure

We probed each component of the final chain checkpoint against every earlier phase checkpoint (deterministic posterior-mode encoding; n=3; pre-registered with the prediction that the reward head would show selective forgetting — a prediction the data refuted):

- *Reward knowledge.* Discrimination D = (mean predicted reward at true success steps) − (at zero-reward steps) on each task's held episodes. The retention ratio $R = D_{\text{final}}/D_{\text{own-phase}}$ for the always-forgotten task: **0.99/1.06/1.01**. The reward head never forgets; on one seed it improves with continued replay.
- *Off-distribution check.* The same discrimination on the degraded policy's own final-era episodes — the states a drifted actor actually visits, including outright failures: $D = 0.87$ (vs. 0.90 on curated episodes), with zero false reward on failure trajectories. The signal is intact where re-teaching needs it, not just where replay polishes it.
- *Values.* Critic means on old-task states remain high and rise across phases (0.84 → 0.91).
- *Termination model.* Continuation-head discrimination at true terminal steps: 0.95–1.0 at every checkpoint, including immediately after the shortest task phase.
- *Representations, with a caveat that matters.* A frozen-action-margin probe shows old-task action preferences under re-encoded latents degrade (margin −0.25 at run end vs. +4.5 and +2.7 for stabler tasks): the latent space *does* drift. Precisely: co-trained heads track the drift; frozen heads decay under it; and the actor — co-trained, but through the RL channel — decays too. "The world model remembers" means replay-maintained knowledge, not a frozen latent geometry.

A scope note: this is *replay-maintained* memory. Without replay, representation overwrite is total (§4.2). The claim is regime-specific and consistent with [6] (§2). In the regime the continual-MBRL literature operates in, the premise "the world model is what needs protecting" has it backwards: the world model is the component replay already protects.

4.4 The isolation reference

For completeness we built the multi-head upper bound made task-agnostic: frozen per-task actor snapshots plus a nearest-centroid task router over encoder embeddings (no task labels at evaluation). With correct routing (measured ∼1.00 per task offline), the oracle read passes 5/5 seeds — but the honest routed read passes 3/5, entirely because the hardest task's *stored* policy delivers only 0.62 ± 0.13 against a 0.6 bar; two seeds flipped verdicts in opposite directions between two reads of the same trained agent. Storing policies caps retention at snapshot quality and converts the pass/fail question into a coin flip at the bar. Isolation is a reference point, not a fix. (This experiment also surfaced an episode-ordering bug in our router evaluation, whose diagnosis — an offline exact replication of the corrupted metric — set the pattern for the selection gauges of §7; the corrected re-read is the number reported here, with both reads in the artifact.)

5 Recovery from the World Model Alone

If §4 is right, the actor should be re-teachable from the world model alone. We froze the entire final world model of each four-task run (encoder, RSSM, all heads; parameter checksums asserted unchanged), selected the always-forgotten task's episodes from the buffer by phase window (selection purity ≥80%, verified by routing), and re-trained only the drifted live actor on imagined rollouts from those states. Two teachers, same starts, same budget (20k updates), zero new environment steps, real-environment evaluation every 500 updates (Figure 1):

- **RL-in-imagination** (the standard DreamerV3 actor–critic update on the frozen model): **0/3**. One seed reaches 0.84 and oscillates back to 0.38; one collapses to 0.0 *and destroys collateral tasks' behavior* (a co-trained task falls to 0.08); one never exceeds 0.57.
- **Dream self-imitation** (roll the current actor with sampling in imagination; grade each trajectory with the frozen reward and value heads; behavior-clone the top 25%): **3/3**, at 2,000/3,500/7,500 updates — 0.38→0.85, 0.66→0.92, 0.66→0.85 — above the frozen-head storage ceiling of §4.4, with collateral tasks intact.

Same model, same knowledge, same dreams. The supervised channel converts the signal into behavior; the policy-gradient channel thrashes. This was the pre-registered arm comparison's strongest row ("only imitation passes → the instability is the RL channel"), and it is why we frame forgetting in this regime as a channel problem. The world model is a sufficient behavioral memory at this scale; what was missing was a stable way to read behavior back out of it.

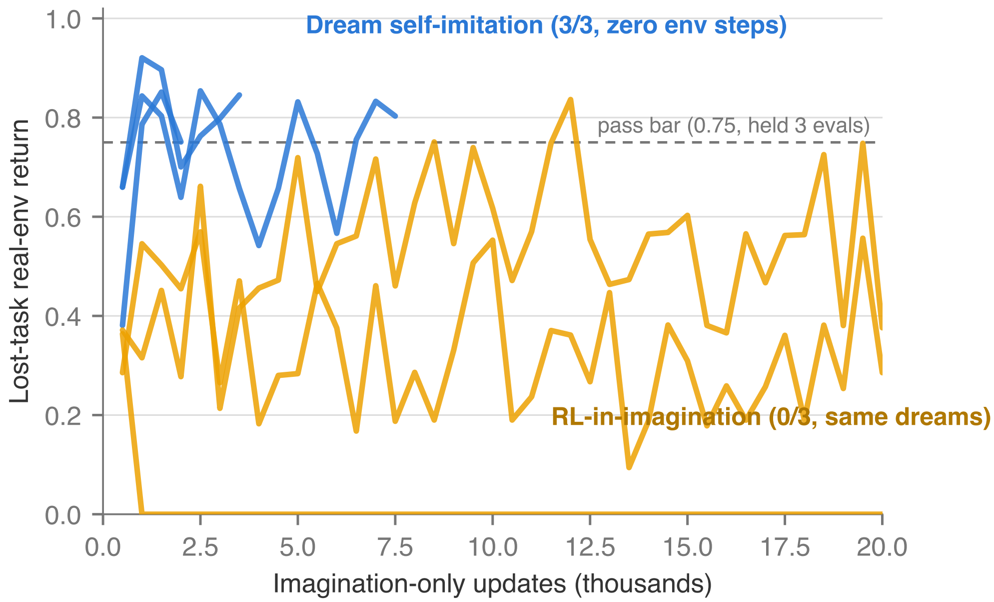

Figure 1: **The recovery race.** Real-environment return on the forgotten task during imagination-only re-teaching; three seeds per arm, identical frozen world model and imagined data. Supervised self-imitation on graded dreams (blue) clears the 0.75 recovery bar on 3/3 seeds within 2,000–7,500 updates; RL-in-imagination (orange) thrashes for 20,000 updates and passes 0/3, with one seed collapsing to zero. Returns are raw environment return in [0, 1]; both arms use the same frozen world model and identical imagined data, so the failure of the orange arm is attributable to the RL channel alone.

6 Dream Rehearsal

The recovery result suggests the fix: never let the actor drift in the first place (Algorithm 1). After each task's phase ends, its buffered episodes are kept as *rehearsal starts*; from then on, after every 2,000 environment steps of new-task training, the agent runs 50 dream-self-imitation updates per prior task — imagine from that task's states with the current (sampling) actor, grade each imagined trajectory with the live reward/continuation/value heads (§7), and behavior-clone the top 25%. One live actor throughout. No task labels at training or evaluation, no frozen policies, no router, no new parameters; the world model and critic train exactly as in the plain recipe. Rehearsal adds ∼15% compute at four tasks (linear in task count).

6.1 Four-task result

3/3 seeds pass all four tasks (plain replay 0/3; isolation reference 3/5); per-seed final retention in Table 1, method comparison in Figure 2.

Seed	DoorKey	SimpleCrossing	LavaGap	MultiRoom	Env. steps
1	0.959	0.905	0.760	0.814	136k
2	0.956	0.740	0.662	0.806	244k
3	0.958	0.826	0.727	0.799	130k

Table 1: Four-task chain, initial-scorer run. §7's corrected grader lifts the one unstable task's post-acquisition stability from 27/19/42% to 95/100/67% with all-task passes preserved on all seeds; corrected-run per-seed means are 0.960/0.868/0.899/0.751, 0.961/0.751/0.944/0.800, and 0.964/0.825/0.784/0.768 (used in Figure 2).

Algorithm 1 Dream rehearsal (one training chunk)

Require: world model W, actor π, critic V; never-clear buffer with per-phase episode libraries $L_1, \ldots, L_{k-1}$ for prior tasks; chunk size C=2000; updates per prior task K=50; clone fraction q=0.25; horizon H=15

train (W, V, π) for C environment steps of the current task, exactly as in the plain recipe
for $j = 1$ **to** $k - 1$ **do**
  for K updates **do**
    sample a batch of states s from L_j; encode with W
    imagine H steps from s with the sampling actor π
    score each imagined trajectory with the realized-first rule (Eq. 1)
    behavior-clone π on the top-q fraction of trajectories
  end for
end for

The historically doomed task (SimpleCrossing) averages **0.824** — above the isolation ceiling's 0.62 ± 0.13 — and stays above bar in **153/160 (96%)** of all post-acquisition evaluations. Two of three seeds finish the whole chain *faster* than the isolation architecture despite the rehearsal overhead, because no skill ever needs relearning: acquisition of later tasks accelerates (LavaGap learns in 8–10k steps in-chain on every seed). Retention converges to each task's own competence ceiling rather than a common value — which is what not forgetting should look like.

6.2 Why dream at all? Comparison against real-episode cloning

The buffer already stores the real episodes, actions included; cloning them directly (CLEAR-style) is the obvious simpler alternative. We pre-registered this comparison adversarially — two of the four interpretation rows favor the simpler method, and publication of the outcome in the main body was committed either way — with the strongest fair version: clone only *competent* episodes (return > 0.05), matched update-for-update to the dream schedule. A methods note: our first implementation of the cloning arm silently misaligned features and actions by one step (the RSSM's action convention lags the actor's decision), which *destroyed* skills while cloning their own successes — an artifact flattering to our method, caught by an isolation gauge (cloning a skill's own success must preserve it) and fixed before any comparison was read. With the corrected update (Table 2):

Arm	T_1 per seed	Mean	All-four passes
Dream rehearsal	0.868 / 0.751 / 0.825	**0.815**	3/3
Real-episode cloning (competent-filtered)	0.630 / 0.678 / 0.743	0.684	3/3

Table 2: Dream rehearsal vs. matched real-episode cloning; T_1 (SimpleCrossing) final retention.

Real-episode cloning *works*: it clears every bar and is cheaper per update, and we report it as a viable baseline. But it retains consistently less: paired difference $+\mathbf{0.131}$, bootstrap 95% CI $[\mathbf{0.073}, \mathbf{0.238}]$ (10k draws), meeting the pre-registered distinctness criterion (CI excludes zero, point estimate ≥0.10) — with complete seed separation (every dream seed exceeds every cloning seed). Imagination contributes beyond the supervised channel; the natural candidates are fresh on-policy state coverage from the current world model, and the gradability of imagined data (§7). The unfiltered cloning variant was not re-run after the bug fix: it clones a strict superset of the filtered arm's data plus pre-competence flailing, and cannot plausibly exceed the filtered arm it would need to beat; we disclose it as unrun.

7 Grading Imagined Trajectories Is Load-Bearing

One systematic instability survived the four-task result: the lethal short-episode task (LavaGap) wobbled after its phase — above bar at the final read on all seeds, but clearing the bar in only 27%/19%/42% of post-acquisition evaluations. Pre-registered probes (predictions committed; two of three refuted) convicted neither the rehearsal data (that task's library was the largest and most competent of all tasks) nor the world model's physics (termination

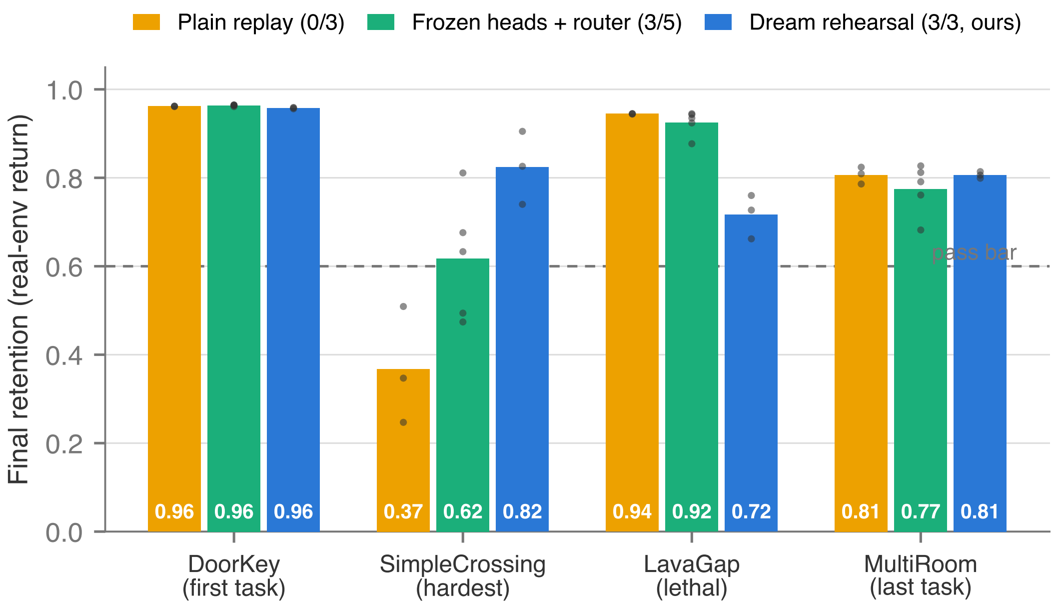

Figure 2: **Final retention across the four-task chain**, three methods, per-seed points over bar means, pass bar 0.6. Plain never-clear replay (orange) holds the easy tasks but collapses on the hardest (SimpleCrossing, 0.37) — 0/3 seeds pass all four. The frozen-heads+router isolation reference (green) passes 3/5. Dream rehearsal (blue) passes 3/3 with the hardest task at 0.82.

discrimination 0.95–1.0 from the earliest checkpoint), but the **scorer**: with ~10-step episodes and a 15-step imagination horizon, **99.7%** of imagined trajectories terminate in-horizon, and a naive score (discounted reward plus terminal value bootstrap) keeps scoring imagination *past the dream's own ending* — post-terminal latents the model never trained on. Selection was near-random on this task (AUC 0.49–0.56): 10–38% of what rehearsal cloned were trajectories that walked into lava, ranked there by post-terminal bootstrap noise. On long-horizon tasks, where dreams rarely terminate in-horizon, the same scorer is clean — which is exactly why only the short-horizon task wobbled.

Fixing this exposed a second, opposite failure mode, caught *offline, before any environment cost*, by a selection gauge we now consider part of the method: generate sibling dreams from banked starts, label each by its own imagined outcome, and measure each candidate scorer's selection quality directly (AUC, top-quartile purity) rather than inferring it from returns. Survival-weighting the rewards (with or without the exact λ-return construction) *inverts* selection on long-horizon tasks (AUC 0.004): a dream that actually reaches the goal has its terminal reward and bootstrap suppressed, while a dream that merely wanders keeps the critic's optimistic value — promises outbid realized successes. A recovery-referee run confirmed the inversion behaviorally (flat 0.0 where the naive scorer had recovered the skill).

The rule that closes both failure modes is **realized-first grading**. Let c_t be the continuation head's probability at imagined step t, $p_t = \prod_{k<t} c_k$ the probability the dream has *reached* step t (so nothing counts past termination), r_t the predicted reward, and V the critic. With discount γ and horizon H:

$$\text{realized} = \sum_{t=0}^{H-1} \gamma^t p_t r_t, \qquad \text{score} = 10 \cdot \mathbf{1}[\text{realized} > 0.3] + \text{realized} + \gamma^H p_H V(s_H). \tag{1}$$

Trajectories that actually achieved reward in imagination outrank all value promises; the termination-aware score orders within groups. The principle is not new on real data — it is the self-imitation advantage gate of Oh et al. [5], $(R - V(s))_+$: imitate only when realized return beats the critic's estimate. Our contribution is scoped to making it safe for imagination, where the critic's optimism is unconstrained by any real outcome and where dreams, unlike real episodes, do not stop at their own endings: the reach-probability weighting, the termination-aware bootstrap, and the offline gauge. On the gauge, realized-first scores AUC 1.0 and top-quartile purity 1.0 on both task profiles; in the recovery referee it passes at 2,000 updates (1.75× faster than the naive scorer); re-running the four-task chain with it lifts the wobbling task's stability to **95%/100%/67%** with every guardrail held (all-four passes on all seeds; hardest-task mean 0.815). We note one adjacent null in the literature that our target distinction predicts: reward-weighted *episode selection for world-model training* does not help and can cost plasticity [1, app. D.8]. Selecting which real episodes train the world model is a different object from selecting which imagined trajectories the actor clones; our gauge measures the latter directly.

8 Scaling to Eight Tasks

Doubling chain length with four audition-gated additional tasks (each verified learnable solo within the phase budget; candidates that could not bootstrap under greedy exploration — long-horizon sparse tasks — were excluded and are reported as the acquisition frontier of this study): DoorKey-5x5 → SimpleCrossing → LavaGap → MultiRoom → LavaCrossing → DistShift2 → DoorKey-6x6 → Unlock. Same recipe, same constants, one changed variable (task count). **All three seeds pass all eight tasks**, exceeding the pre-registered ≥2/3 bar (Table 3).

Seed	DoorKey5	SimpleX	LavaGap	MultiRoom	LavaCross	DistShift	DoorKey6	Unlock
1	0.964	0.943	0.947	0.759	0.861	0.960	0.915	0.861
2	0.963	0.919	0.943	0.746	0.865	0.929	0.962	0.888
3	0.901	0.876	0.942	0.754	0.794	0.959	0.962	0.740

Table 3: Eight-task chain, per-seed final retention; pass bar 0.6 on every task.

By the final phase, rehearsal runs 350 dream-imitation updates per 2,000-step chunk across seven prior tasks — still one actor, still no task labels, still no added parameters.

Two secondary reads matter more than the pass itself. **Consistency:** per-task cross-seed ranges are ≤0.07 on seven of eight tasks (the exception is Unlock, range 0.15, still entirely above bar). The isolation reference of §4.4 delivered 0.62 ± 0.13 on its hardest task at *half* this chain length, with two seeds flipping verdicts between two reads of the same agent. At twice the tasks, rehearsal is materially more stable than isolation was at half. **The predicted scale-failure did not appear:** we pre-registered rehearsal dilution (early tasks sagging as a fixed per-task budget spreads over more tasks) as the expected failure mode, with a stability-weighted allocation rule as its remedy. No dilution is observed — the incumbent quartet holds at or above its four-task band — so the remedy is not triggered and we do not build it.

Uncontrolled observation, explicitly not a result: the historically doomed task reads higher at eight tasks (0.943/0.919/0.876) than at four (0.868/0.751/0.825). We decline to interpret this: the gap between arm means is the same magnitude as the four-task arm's own seed-to-seed spread, the comparison crosses two different experiments (different downstream tasks, rehearsal-pass counts, and budgets), and no matched-budget control was run. It is recorded only so that a future controlled test — same chain, varying only downstream task count — has a reason to exist.

Retention again settles at each task's own ceiling rather than a common value (easy tasks ~0.96; MultiRoom ~0.75, its band in every chain we have run; LavaCrossing ~0.86).

9 Limitations

MiniGrid scale; a 17M-parameter world model; one task ordering per chain length; return bars rather than normalized scores (we additionally report retention relative to each task's own post-acquisition competence). n=3 with real run-level nondeterminism — identical seeds can diverge qualitatively — so we report dispersion and bar-distance, not only pass counts. We acknowledge n=3 is below the 5–10 common at scale; single-workstation compute bounds it, and we compensate by reporting complete per-seed values for every headline comparison, several of which (§5, §6.2) show complete seed separation, which no aggregation can manufacture. Rehearsal cost grows linearly with task count; the observed acquisition speed-up offsets it at n=8, but the crossover point is unmeasured. The buffer retains raw experience: dream rehearsal removes stored *policies*, not stored *data* — replacing rehearsal starts with generated or bounded starts is future work. Tasks that cannot bootstrap under greedy exploration within our budget mark the acquisition frontier of this study; nothing here addresses exploration. Two scoring bugs shipped during development and were caught by our own offline gauges before contaminating results; we report both (§4.4, §7) and draw the methodological conclusion that selection gauges belong in the method, not the appendix.

10 Conclusion

In replay-maintained world-model agents, forgetting is not a memory problem: the world model retains what old tasks need, and the actor loses the behavior anyway. Once localized, the failure has a direct fix — read behavior back out of the world model through a supervised channel, continuously, with graded dreams — and the fix holds unanimously through eight-task chains on a single workstation GPU. The grading rule, not the rehearsal schedule, is where the difficulty lives; we ship the gauge that measures it. If the pattern holds beyond gridworlds, the implication for continual MBRL is a reframing: protect less, and re-teach from what the model already knows.

Reproducibility statement

Every experiment was pre-registered: protocol, bars, and an interpretation matrix committed to version control before launch (commit hashes in the artifact index), with off-machine timestamped archives. All refuted hypotheses are reported. Code, run summaries, probe artifacts, per-round evaluation traces, and the full pre-registration trail are available at `https://github.com/gurpnijjer/dream-rehearsal` (archived at doi:10.5281/zenodo.21462836). Total compute: a single NVIDIA GB10 workstation; the complete experimental record of this paper is approximately three weeks of one GPU.

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A Protocol Details and Hyperparameters

Observation	$64 \times 64 \times 3$ RGB, partial egocentric view
World model	17M parameters (CNN encoder, RSSM, reward/continuation/decoder heads)
Actor	1.8M parameters; one-hot action distribution
Imagination horizon	$H = 15$
Episode time limit	256 environment steps
Train ratio	512 (reference-implementation semantics)
Evaluation	every 2,000 env. steps; 10 real episodes per learned task
Phase advance	all learned tasks ≥ 0.6 mean return for 3 consecutive rounds
Phase cap	150k env. steps
Final retention	mean of last 3 evaluation rounds per task
Rehearsal schedule	50 dream-imitation updates per prior task per 2,000-step chunk
Clone fraction	top 25% of imagined trajectories per batch
Realized threshold	0.3 (Eq. 1)
Exploration	greedy (no curiosity bonus)
Seeds	1, 2, 3 for every graded experiment; no per-seed tuning

Table 4: Protocol constants. All remaining hyperparameters are the reference DreamerV3 MiniGrid defaults; the exact configuration files ship in the repository.

Pre-registration. Each experiment's protocol document (bars, arms, interpretation matrix, and predicted failure modes) was committed to version control before launch; amendments after first data are labeled as such in the released trail, with the triggering observation disclosed. The released artifact includes every protocol, every amendment, and both scoring-bug post-mortems.

B Per-Seed Data

All per-run summaries (`chain_summary.json`) and full evaluation traces (`chain_metrics.jsonl`; one row per evaluation round per task) for every run in this paper — plain-replay chains, freeze brackets, isolation-reference runs, recovery races, both four-task rehearsal runs (initial and corrected scorer), the cloning-comparison arms, and the eight-task chains — are included in the released repository under `results/`.

C Candidate Mechanisms for the RL Channel's Instability

Section 5 establishes *that* the policy-gradient channel fails where supervised imitation succeeds under identical conditions, not *why*. We record the candidate explanations consistent with our observations, explicitly as untested hypotheses. **(1) Critic-optimism contamination.** §7 documents that in imagination the critic's optimism is unconstrained by real outcomes (value promises outbid realized successes when scoring trajectories). The same optimism enters the actor–critic objective directly through the bootstrapped return, so the RL channel chases promises no realized dream validates — consistent with the observed oscillation without convergence (one seed reaching 0.84 and returning to 0.38). The supervised channel's targets are realized actions of graded trajectories, with no bootstrap term. **(2) Update-magnitude interference.** One RL seed destroyed a co-trained task's behavior (0.08) through the shared actor; the supervised channel produced no collateral damage in any run. This is consistent with high-variance policy-gradient updates propagating broadly through shared parameters. **(3) Relation to plasticity findings.** Capacity loss [12] and primacy bias [13] document RL-channel degradation under nonstationarity during ordinary training; our recovery setting shows failure even with a stationary, fully trained world model, isolating the channel from the representation. Adjudicating among these requires interventions we have not run — variance clipping, critic-pessimism ablations, entropy schedules — and we list them as the natural next experiments.